

Lab 4 : NER on Scientific Texts

Objective : Build a system to automatically identify and classify Named Entities in scientific texts (e.g., methods, datasets, software, researchers, institutions).

Steps :

1. **Text extraction :** Extract the text from a collection of scientific papers in PDF format using tools such as `pdfminer`, `PyPDF`, or `Grobid`.
2. **Text segmentation :** Divide the document into logical sections such as :
 - Abstract
 - Introduction
 - Related Work
 - Methodology
 - Experiments / Results
3. **Cleaning :** Remove irrelevant or noisy elements such as :
 - References and bibliography
 - Citations (e.g., [12], (Smith, 2021))
 - Figure and table captions
 - Mathematical formulas
4. **Text preprocessing :**
 - Tokenize the text into sentences and words.
 - Normalize the text (lowercasing when necessary).
 - Remove stopwords if required.
5. **Named Entity Recognition (NER) :** Apply an NER model (using libraries such as `spaCy`, `Flair`, or `HuggingFace Transformers`) to detect entities such as :
 - Person
 - Organization
 - Location
 - Method / Algorithm
 - Dataset
 - Software / Tools
6. **Post-processing and filtering :**
 - Remove duplicated entities.
 - Normalize entity names (e.g., CNN and Convolutional Neural Network).
7. **Entity analysis :**
 - Identify the most frequent entities in each section.

- Determine which methods, datasets, and tools are most cited.
- Analyze the distribution of entity types across sections.

8. **Visualization :**

- Create bar charts showing the frequency of entity types.
- Generate a word cloud of the most frequent entities.
- Build a network graph linking methods and datasets.